

**Enhanced Detection of Age-Related and Cognitive Declines Using
Automated Hippocampal-to-Ventricle Ratio in Alzheimer's
Patients**

Supplementary material

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Supplementary Tables

	Hippocampus		Ventricles	
	Left	Right	Left	Right
MALF				
Accuracy	1.00 (0.00)	1.00 (0.00)	1.00 (0.00)	1.00 (0.00)
Dice	0.90 (0.03)	0.90 (0.04)	0.75 (0.06)	0.73 (0.06)
Kappa	0.91 (0.03)	0.92 (0.03)	0.71 (0.08)	0.68 (0.09)
Sensitivity	0.91 (0.03)	0.92 (0.03)	0.71 (0.08)	0.69 (0.09)
Specificity	1.00 (0.00)	1.00 (0.00)	1.00 (0.00)	1.00 (0.00)
NLPB				
Accuracy	1.00 (0.00)	1.00 (0.00)	1.00 (0.00)	1.00 (0.00)
Dice	0.90 (0.01)	0.90 (0.02)	0.73 (0.05)	0.72 (0.05)
Kappa	0.90 (0.03)	0.90 (0.03)	0.70 (0.07)	0.68 (0.07)
Sensitivity	0.90 (0.03)	0.90 (0.03)	0.70 (0.07)	0.68 (0.07)
Specificity	1.00 (0.00)	1.00 (0.00)	1.00 (0.00)	1.00 (0.00)
CNN				
Accuracy	1.00 (0.00)	1.00 (0.00)	1.00 (0.00)	1.00 (0.00)
Dice	0.96 (0.01)	0.96 (0.01)	0.91 (0.04)	0.89 (0.04)
Kappa	0.95 (0.01)	0.95 (0.02)	0.89 (0.05)	0.87 (0.05)
Sensitivity	0.95 (0.01)	0.95 (0.02)	0.89 (0.05)	0.87 (0.05)
Specificity	1.00 (0.00)	1.00 (0.00)	1.00 (0.00)	1.00 (0.00)

Median (SD).

TABLE S1 – Automatic segmentation performance metrics.

Performance metrics of the three cross-validation trained automated segmentation methods (MALF, NLPB, and CNN) for both the hippocampus (HCvol) and the surrounding ventricular regions (CSFvol). Overall-agreement and class-wise performance metrics include accuracy, Dice coefficient, Cohen’s Kappa, sensitivity and specificity, with separate values reported for the left and right hemispheres. All values are reported as medians with standard deviations.

	F	df	p
Hippocampus			
Accuracy	25.77742	(1.65, 77.40)	0.000
Dice	513.43885	(3.00, 141.00)	0.000
Kappa	127.90027	(3.65, 171.33)	0.000
Sensitivity	128.04824	(3.65, 171.39)	0.000
Specificity	32.43853	(4.10, 192.77)	0.000
Ventricles			
Accuracy	25.38914	(3.81, 178.98)	0.000
Dice	644.56452	(3.73, 175.12)	0.000
Kappa	249.27515	(3.33, 156.29)	0.000
Sensitivity	249.28824	(3.33, 156.30)	0.000
Specificity	11.02206	(3.88, 182.54)	0.000

Repeated-measures robust two-way ANOVA with trimmed means. p values controlled for false discovery rate based on Benjamini & Hochberg.

TABLE S2 – Two-way ANOVA results for segmentation metrics.

Results of the robust repeated-measures two-way ANOVA with trimmed means applied to the hippocampal and ventricular segmentation performance measures comparing the automated segmentation methods and hemispheres. Reported are the F-statistics, degrees of freedom, and p -values. P -values have been adjusted for FDR using the Benjamini and Hochberg procedure.

	Hippocampus			Ventricles		
	ψ	95% CI	p	ψ	95% CI	p
MALF						
Accuracy	0	(0, 0)	-	0	(0, 0)	-
Dice	-2.2e-03	(-9.5e-03, 5.1e-03)	0.362	0.02	(1.4e-03, 0.03)	0.001
Kappa	-8.4e-03	(-0.02, 1.1e-03)	0.009	0.02	(-4.0e-03, 0.04)	0.017
Sensitivity	-8.4e-03	(-0.02, 1.1e-03)	0.009	0.02	(-4.0e-03, 0.04)	0.017
Specificity	0	(0, 0)	-	0	(0, 0)	-
NLPB						
Accuracy	0	(0, 0)	-	0	(0, 0)	-
Dice	1.2e-03	(-2.7e-03, 5.2e-03)	0.332	0.02	(5.3e-03, 0.03)	0.000
Kappa	2.3e-03	(-6.7e-03, 0.01)	0.431	0.02	(2.8e-04, 0.04)	0.003
Sensitivity	2.3e-03	(-6.7e-03, 0.01)	0.431	0.02	(2.4e-04, 0.04)	0.003
Specificity	0	(0, 0)	-	0	(0, 0)	-
CNN						
Accuracy	0	(0, 0)	-	1.5e-05	(-1.2e-05, 4.1e-05)	0.092
Dice	1.5e-03	(-5.9e-04, 3.5e-03)	0.032	0.02	(8.6e-03, 0.02)	0.000
Kappa	2.2e-04	(-3.0e-03, 3.4e-03)	0.832	0.02	(0.01, 0.03)	0.000
Sensitivity	2.2e-04	(-3.0e-03, 3.4e-03)	0.832	0.02	(0.01, 0.03)	0.000
Specificity	0	(0, 0)	-	0	(0, 0)	-

Robust pair-wise posthoc comparisons using trimmed means. Bold cells show significance after multiple comparison correction.

TABLE S3 – Pair-wise comparisons for segmentation metrics by side.

Robust pair-wise post hoc comparisons of side differences in performance measures within the automated method for hippocampal and ventricular segmentations. The comparisons include robust effect sizes (ψ) with their 95% confidence intervals and corresponding p -values. Bold entries indicate statistical significance after correction for multiple comparisons.

	Hippocampus			Ventricles		
	ψ	95% CI	p	ψ	95% CI	p
Accuracy						
MALF-L — CNN-L	-7.3e-05	(-1.0e-04, -4.5e-05)	0.000	-9.6e-05	(-1.2e-04, -7.2e-05)	0.000
MALF-L — CNN-R	-5.4e-05	(-8.3e-05, -2.5e-05)	0.000	-6.5e-05	(-9.3e-05, -3.6e-05)	0.000
MALF-R — CNN-L	-6.5e-05	(-9.3e-05, -3.6e-05)	0.000	-8.5e-05	(-1.1e-04, -5.9e-05)	0.000
MALF-R — CNN-R	-4.8e-05	(-7.7e-05, -1.9e-05)	0.000	-5.4e-05	(-8.3e-05, -2.5e-05)	0.000
NLPB-L — CNN-L	-6.5e-05	(-9.3e-05, -3.6e-05)	0.000	-9.4e-05	(-1.2e-04, -6.9e-05)	0.000
NLPB-L — CNN-R	-4.4e-05	(-7.3e-05, -1.5e-05)	0.000	-6.9e-05	(-9.7e-05, -4.0e-05)	0.000
NLPB-R — CNN-L	-6.7e-05	(-9.5e-05, -3.8e-05)	0.000	-9.2e-05	(-1.2e-04, -6.7e-05)	0.000
NLPB-R — CNN-R	-5.0e-05	(-7.9e-05, -2.1e-05)	0.000	-6.9e-05	(-9.7e-05, -4.0e-05)	0.000
MALF — NLPB	0	(0, 0)	-	0	(0, 0)	-
Dice						
MALF-L — CNN-L	-0.06	(-0.07, -0.05)	0.000	-0.16	(-0.18, -0.15)	0.000
MALF-L — CNN-R	-0.06	(-0.06, -0.05)	0.000	-0.15	(-0.16, -0.13)	0.000
MALF-R — CNN-L	-0.06	(-0.06, -0.05)	0.000	-0.17	(-0.19, -0.16)	0.000
MALF-R — CNN-R	-0.05	(-0.06, -0.05)	0.000	-0.16	(-0.17, -0.14)	0.000
NLPB-L — CNN-L	-0.06	(-0.06, -0.05)	0.000	-0.17	(-0.18, -0.16)	0.000
NLPB-L — CNN-R	-0.06	(-0.06, -0.05)	0.000	-0.15	(-0.17, -0.14)	0.000
NLPB-R — CNN-L	-0.06	(-0.06, -0.05)	0.000	-0.19	(-0.20, -0.17)	0.000
NLPB-R — CNN-R	-0.06	(-0.06, -0.05)	0.000	-0.17	(-0.18, -0.16)	0.000
MALF-L — NLPB-L	-1.6e-03	(-7.3e-03, 4.1e-03)	0.389	8.2e-03	(-7.0e-03, 0.02)	0.103
MALF-L — NLPB-R	-8.1e-04	(-9.0e-03, 7.4e-03)	0.761	0.03	(6.2e-03, 0.05)	0.000
MALF-R — NLPB-L	1.9e-03	(-3.0e-03, 6.8e-03)	0.241	-4.7e-03	(-0.02, 0.01)	0.392
MALF-R — NLPB-R	2.3e-03	(-2.8e-03, 7.3e-03)	0.176	0.01	(-5.6e-04, 0.03)	0.005
Kappa						
MALF-L — CNN-L	-0.05	(-0.06, -0.03)	0.000	-0.17	(-0.20, -0.14)	0.000
MALF-L — CNN-R	-0.05	(-0.06, -0.03)	0.000	-0.15	(-0.18, -0.12)	0.000
MALF-R — CNN-L	-0.04	(-0.05, -0.03)	0.000	-0.19	(-0.22, -0.17)	0.000
MALF-R — CNN-R	-0.04	(-0.05, -0.03)	0.000	-0.16	(-0.19, -0.14)	0.000
NLPB-L — CNN-L	-0.05	(-0.06, -0.04)	0.000	-0.19	(-0.21, -0.17)	0.000
NLPB-L — CNN-R	-0.05	(-0.06, -0.04)	0.000	-0.17	(-0.19, -0.15)	0.000
NLPB-R — CNN-L	-0.06	(-0.06, -0.05)	0.000	-0.21	(-0.23, -0.19)	0.000
NLPB-R — CNN-R	-0.06	(-0.06, -0.05)	0.000	-0.19	(-0.21, -0.17)	0.000
MALF-L — NLPB-L	9.9e-03	(8.2e-04, 0.02)	0.002	0.02	(-0.01, 0.05)	0.107
MALF-L — NLPB-R	0.01	(1.6e-03, 0.02)	0.001	0.03	(4.4e-04, 0.06)	0.003
MALF-R — NLPB-L	0.02	(8.8e-03, 0.03)	0.000	3.5e-03	(-0.03, 0.03)	0.730
MALF-R — NLPB-R	0.02	(0.01, 0.03)	0.000	0.02	(-0.01, 0.05)	0.082

Robust pair-wise posthoc comparisons using trimmed means. Bold cells show significance after multiple comparison correction.

TABLE S4 – Pair-wise comparisons for segmentation metrics by overall-agreement performance.

Robust pair-wise post hoc comparisons of overall-agreement performance (accuracy, Dice coefficient and Cohen’s kappa) between the different automated methods for hippocampal and ventricular segmentations.

The comparisons include robust effect sizes (ψ) with their 95% confidence intervals and corresponding p -values. Bold entries indicate statistical significance after correction for multiple comparisons.

	Hippocampus			Ventricles		
	ψ	95% CI	p	ψ	95% CI	p
Sensitivity						
MALF-L — CNN-L	-0.05	(-0.06, -0.03)	0.000	-0.17	(-0.20, -0.14)	0.000
MALF-L — CNN-R	-0.05	(-0.06, -0.03)	0.000	-0.15	(-0.18, -0.12)	0.000
MALF-R — CNN-L	-0.04	(-0.05, -0.03)	0.000	-0.19	(-0.22, -0.17)	0.000
MALF-R — CNN-R	-0.04	(-0.05, -0.03)	0.000	-0.16	(-0.19, -0.14)	0.000
NLPB-L — CNN-L	-0.05	(-0.06, -0.04)	0.000	-0.19	(-0.21, -0.17)	0.000
NLPB-L — CNN-R	-0.05	(-0.06, -0.04)	0.000	-0.17	(-0.19, -0.15)	0.000
NLPB-R — CNN-L	-0.06	(-0.06, -0.05)	0.000	-0.21	(-0.23, -0.19)	0.000
NLPB-R — CNN-R	-0.06	(-0.06, -0.05)	0.000	-0.19	(-0.21, -0.17)	0.000
MALF-L — NLPB-L	9.9e-03	(8.2e-04, 0.02)	0.001	0.02	(-0.01, 0.05)	0.107
MALF-L — NLPB-R	0.01	(1.6e-03, 0.02)	0.001	0.03	(4.3e-04, 0.06)	0.003
MALF-R — NLPB-L	0.02	(8.8e-03, 0.03)	0.000	3.5e-03	(-0.03, 0.03)	0.732
MALF-R — NLPB-R	0.02	(0.01, 0.03)	0.000	0.02	(-0.01, 0.05)	0.082
Specificity						
MALF-L — CNN-L	-7.9e-05	(-1.1e-04, -5.2e-05)	0.000	-4.6e-05	(-7.5e-05, -1.7e-05)	0.000
MALF-L — CNN-R	-7.7e-05	(-1.0e-04, -5.0e-05)	0.000	-4.4e-05	(-7.3e-05, -1.5e-05)	0.000
MALF-R — CNN-L	-8.7e-05	(-1.1e-04, -6.2e-05)	0.000	-4.0e-05	(-6.8e-05, -1.1e-05)	0.000
MALF-R — CNN-R	-8.5e-05	(-1.1e-04, -5.9e-05)	0.000	-3.7e-05	(-6.6e-05, -8.8e-06)	0.000
NLPB-L — CNN-L	-6.2e-05	(-9.1e-05, -3.4e-05)	0.000	-4.8e-05	(-7.7e-05, -1.9e-05)	0.000
NLPB-L — CNN-R	-6.0e-05	(-8.9e-05, -3.2e-05)	0.000	-4.8e-05	(-7.7e-05, -1.9e-05)	0.000
NLPB-R — CNN-L	-5.8e-05	(-8.7e-05, -2.9e-05)	0.000	-4.4e-05	(-7.3e-05, -1.5e-05)	0.000
NLPB-R — CNN-R	-5.6e-05	(-8.5e-05, -2.7e-05)	0.000	-4.4e-05	(-7.3e-05, -1.5e-05)	0.000
MALF-L — NLPB-L	-6.2e-06	(-3.1e-05, 1.8e-05)	0.437	0	(0, 0)	-
MALF-L — NLPB-R	-8.3e-06	(-3.3e-05, 1.7e-05)	0.310	0	(-4.0e-05, 4.0e-05)	1.000
MALF-R — NLPB-L	-1.2e-05	(-3.8e-05, 1.3e-05)	0.142	4.2e-06	(-2.0e-05, 2.8e-05)	0.597
MALF-R — NLPB-R	-1.0e-05	(-3.6e-05, 1.5e-05)	0.213	0	(0, 0)	-

Robust pair-wise posthoc comparisons using trimmed means. Bold cells show significance after multiple comparison correction.

TABLE S5 – Pair-wise comparisons for segmentation metrics by class-wise performance.
Robust pair-wise post hoc comparisons of class-wise performance (sensitivity and specificity) between the different automated methods for hippocampal and ventricular segmentations. The comparisons include robust effect sizes (ψ) with their 95% confidence intervals and corresponding p -values. Bold entries indicate statistical significance after correction for multiple comparisons.

	Hippocampus	
	Left	Right
CH, N: 20		
Accuracy	0.99 (0.00)	0.99 (0.00)
Dice	0.91 (0.01)	0.90 (0.02)
Kappa	0.91 (0.02)	0.92 (0.03)
Sensitivity	0.91 (0.02)	0.92 (0.03)
Specificity	1.00 (0.00)	1.00 (0.00)
MCI, N: 20		
Accuracy	0.99 (0.00)	0.99 (0.00)
Dice	0.90 (0.02)	0.91 (0.02)
Kappa	0.88 (0.04)	0.92 (0.03)
Sensitivity	0.88 (0.04)	0.92 (0.03)
Specificity	1.00 (0.00)	1.00 (0.00)
AD, N: 20		
Accuracy	0.99 (0.00)	0.99 (0.00)
Dice	0.89 (0.02)	0.89 (0.02)
Kappa	0.88 (0.04)	0.89 (0.04)
Sensitivity	0.89 (0.03)	0.90 (0.04)
Specificity	1.00 (0.00)	1.00 (0.00)

Median (SD).

TABLE S6 – Out-of-sample validation segmentation performance metrics.
Performance metrics of the segmentation of the hippocampus on the out-of-sample validation ADNI subset of data using our trained CNN. Overall-agreement and class-wise performance metrics include accuracy, Dice coefficient, Cohen’s Kappa, sensitivity and specificity, with separate values reported for the left and right hemispheres and stratified by clinical group (CH, MCI, and AD). All values are reported as medians with standard deviations.

	F	df	p
Group			
Accuracy	3.08	(2, 20.26)	0.339
Dice	3.23	(2, 21.73)	0.297
Kappa	3.00	(2, 21.75)	0.176
Sensitivity	3.14	(2, 21.74)	0.158
Specificity	1.02	(2, 21.14)	0.825
Side			
Accuracy	0.35	(1, 30.27)	0.929
Dice	1.27	(1, 32.23)	0.567
Kappa	8.48	(1, 25.03)	0.037
Sensitivity	8.47	(1, 25.12)	0.037
Specificity	1.88	(1, 30.54)	0.825
Group:Side			
Accuracy	2.03	(2, 21.23)	0.391
Dice	1.13	(2, 21.84)	0.567
Kappa	1.46	(2, 20.20)	0.426
Sensitivity	1.45	(2, 20.30)	0.429
Specificity	0.73	(2, 21.63)	0.825

Robust two-way mixed ANOVA with trimmed means. p values controlled for false discovery rate based on Benjamini & Hochberg.

TABLE S7 – Out-of-sample validation two-way mixed ANOVA results.

Results of the robust mixed two-way ANOVA with trimmed means applied to the hippocampal segmentation performance measures on the out-of-sample ADNI subset of data using the trained CNN, comparing clinical group and hemispheres. Reported are the F-statistics, degrees of freedom, and p -values. P -values have been adjusted for FDR using the Benjamini and Hochberg procedure.

Supplementary Figures

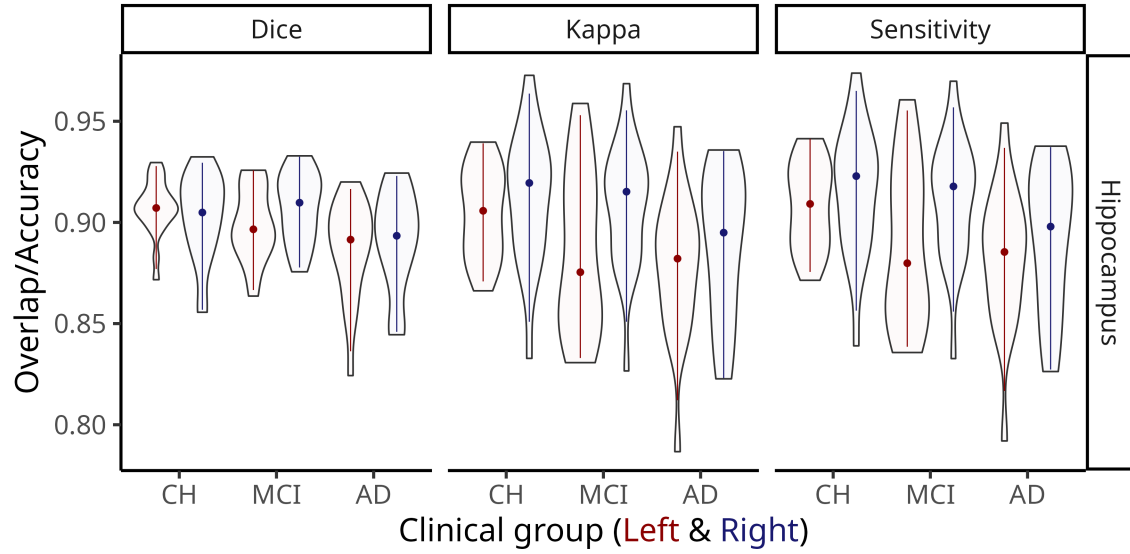


FIGURE S1 – Out-of-sample validation segmentation performance.

Violin plots of the distributions of the overlap and accuracy metrics (Dice, Kappa, and Sensitivity) between manual and computed segmentations of the hippocampus from the left (red) and right (blue) hemispheres on an out-of-sample subset of ADNI data using CNN for each of the clinical groups (CH, MCI, and AD). The median and standard deviation are represented inside the violin plots.

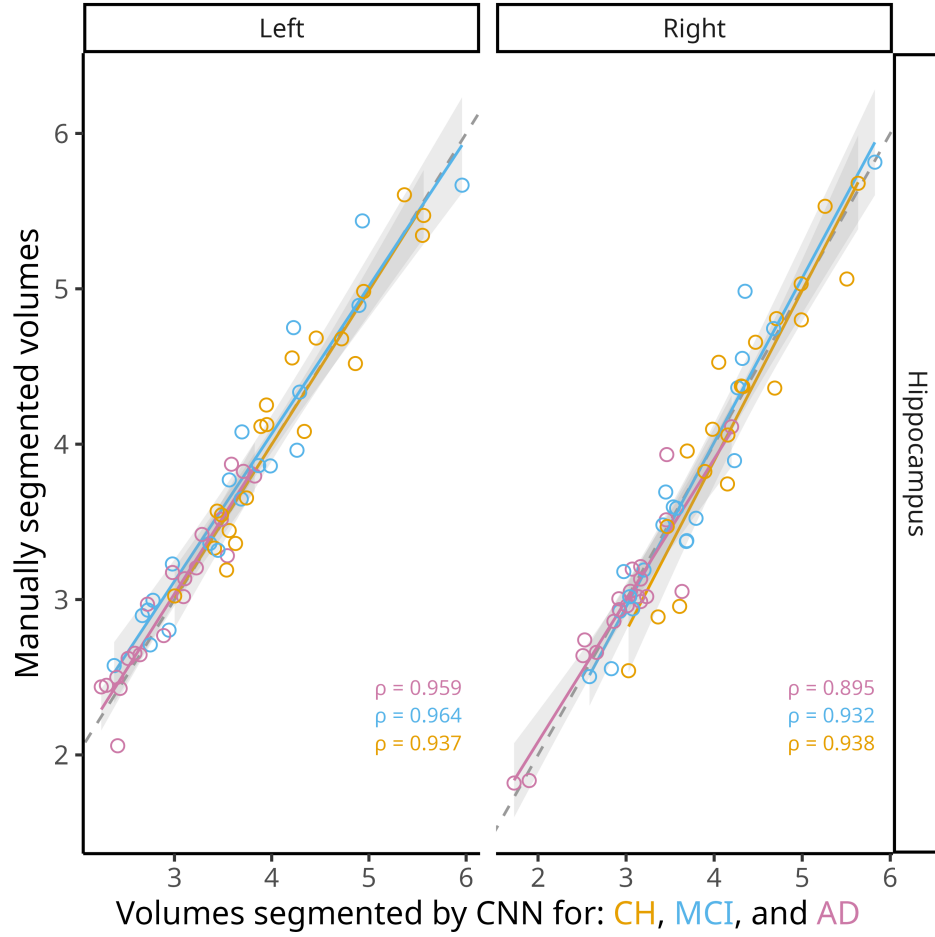


FIGURE S2 – Out-of-sample validation Relationship between manual and computed hippocampal.

Scatter plots of the volumes (measured in cc) of the left and right hippocampus from the manual segmentations against those obtained from CNN on the out-of-sample subset of ADNI data for each of the clinical groups (CH, MCI, AD). The Spearman's rank correlation coefficients (ρ) and their significance are added for each hemisphere.

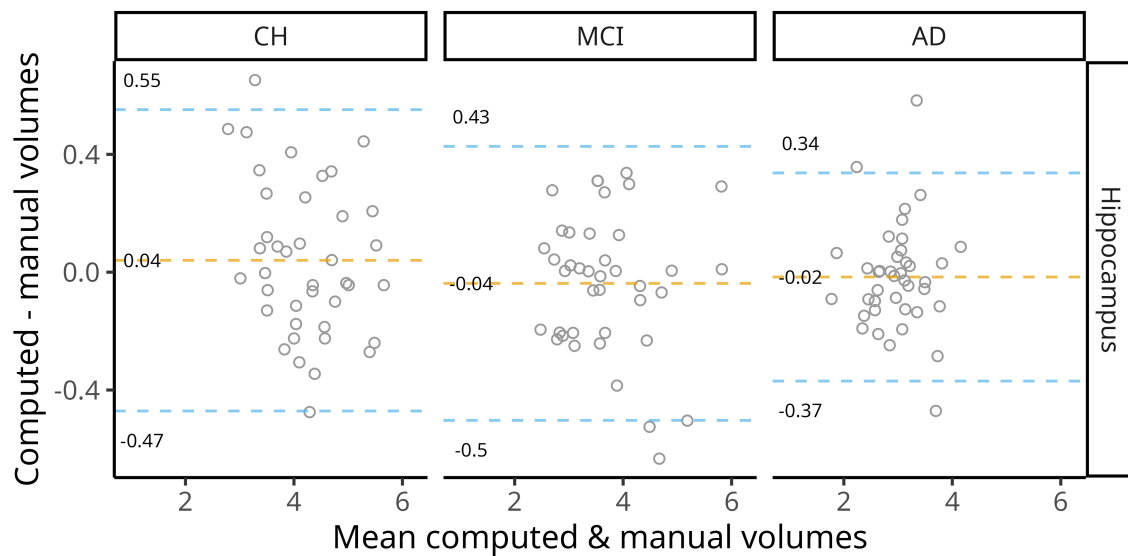


FIGURE S3 – Out-of-sample validation Bland-Altman plots.

Bland-Altman plots for the volumetric comparison between mean HCvol (in cc) resulting from the automatic segmentation of the out-of-sample subset of ADNI data for each of the clinical groups (CH, MCI, AD). The dashed lines represent the mean difference (orange) and the upper and lower limits of agreement (blue).

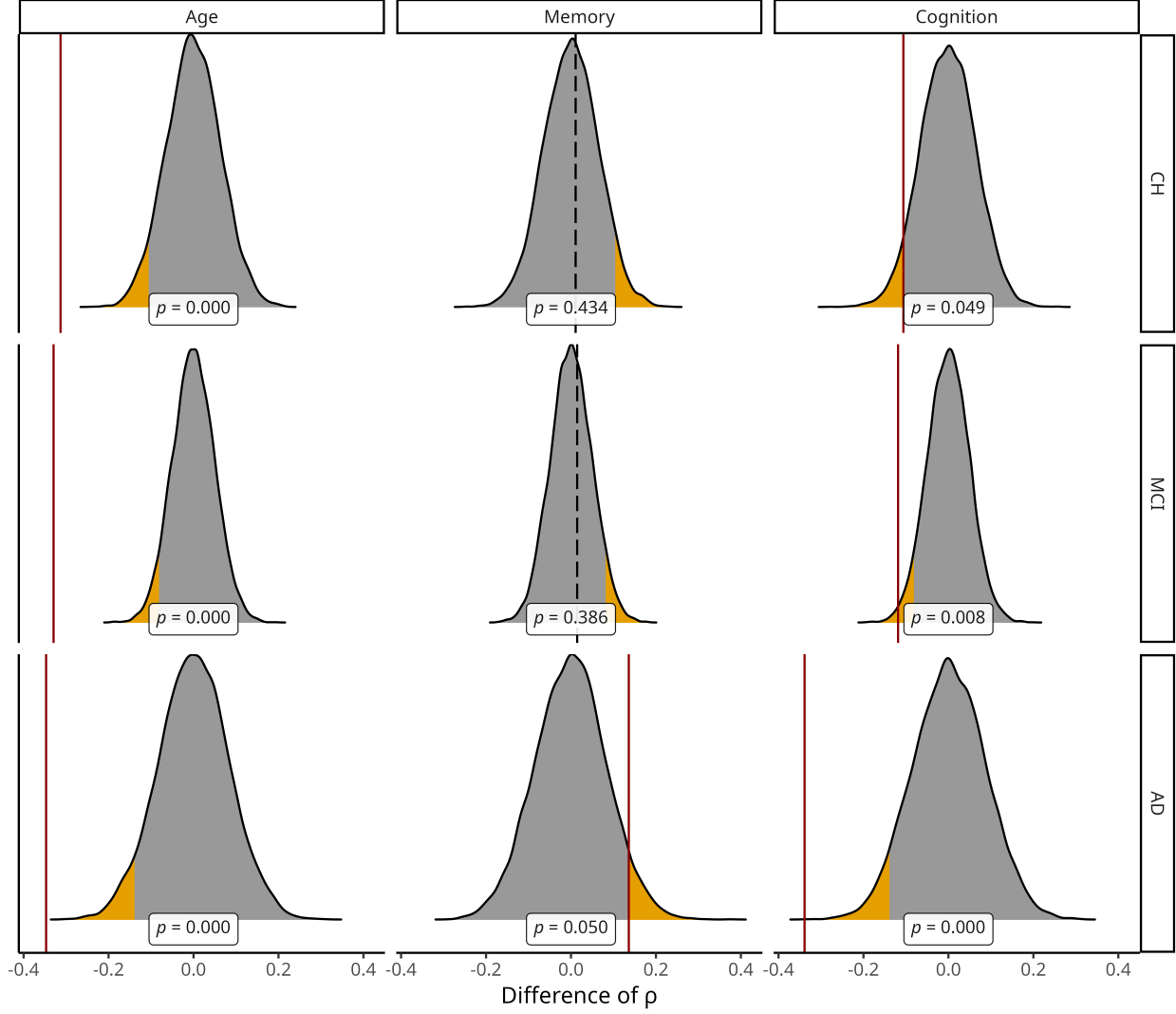


FIGURE S4 – Permutation tests for differences in correlation strength between HCvol and HVR using MALF segmentations.

Distributions of the differences in Spearman's rank correlation coefficients ($\Delta\rho = \rho(\text{HCvol}) - \rho(\text{HVR})$) with respect to age, memory, and global cognition. The hypothesis tested is that the MALF-based HVR measure yields stronger correlations in magnitude than the MALF-based HCvol measure. Each distribution was obtained via 10,000 one-tailed permutation tests; shaded orange area represents the region of rejection of the null hypothesis ($\alpha < 0.05$). Vertical lines mark the observed $\Delta\rho$; solid and red when significant, black and dashed when not. FDR-corrected p -values are provided as labels.

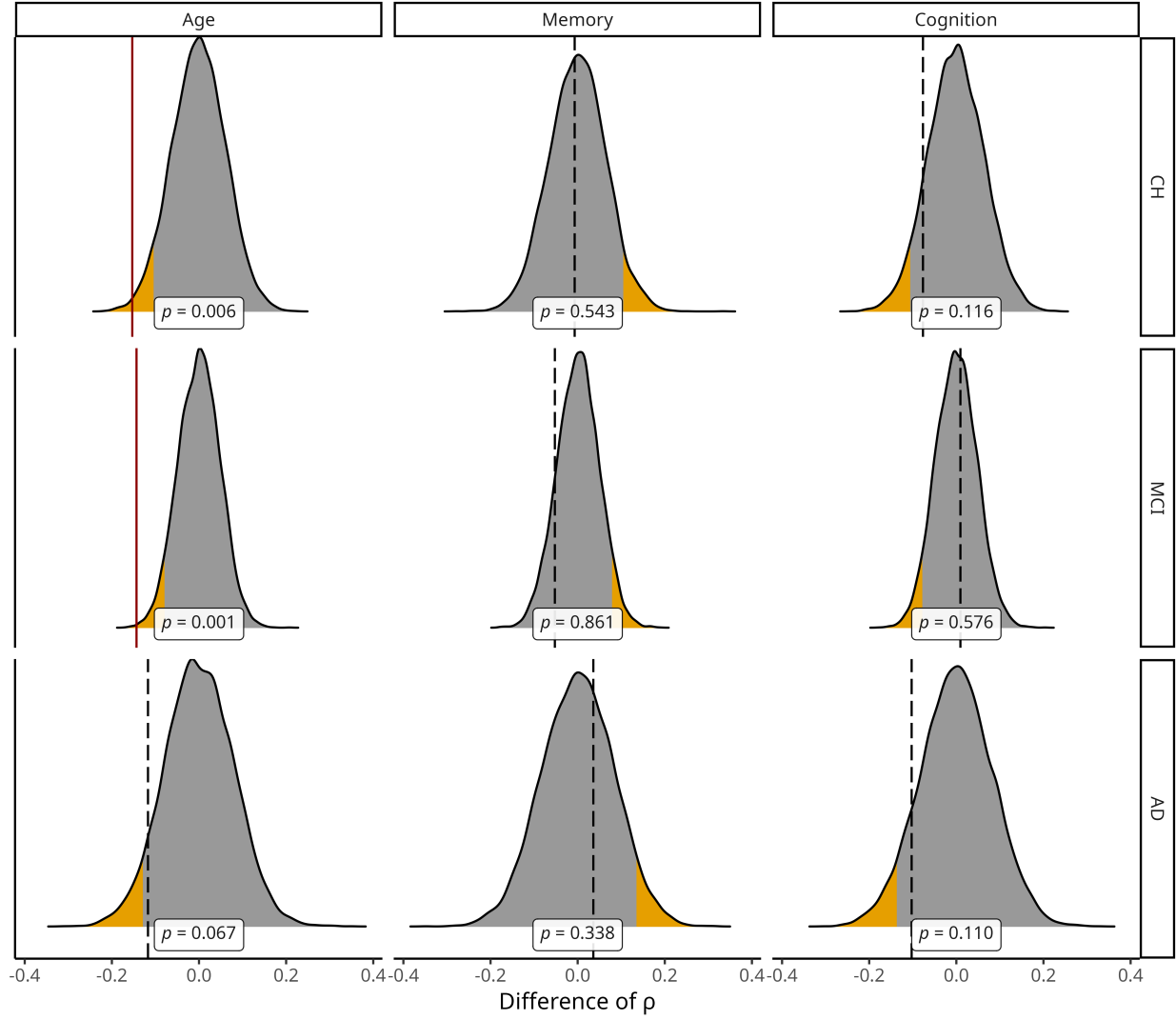


FIGURE S5 – Permutation tests for differences in correlation strength between HCvol and HVR using NLPB segmentations.

Distributions of the differences in Spearman's rank correlation coefficients ($\Delta\rho = \rho(\text{HCvol}) - \rho(\text{HVR})$) with respect to age, memory, and global cognition. The hypothesis tested is that the NLPB-based HVR measure yields stronger correlations in magnitude than the NLPB-based HCvol measure. Each distribution was obtained via 10,000 one-tailed permutation tests; shaded orange area represents the region of rejection of the null hypothesis ($\alpha < 0.05$). Vertical lines mark the observed $\Delta\rho$; solid and red when significant, black and dashed when not. FDR-corrected p -values are provided as labels.

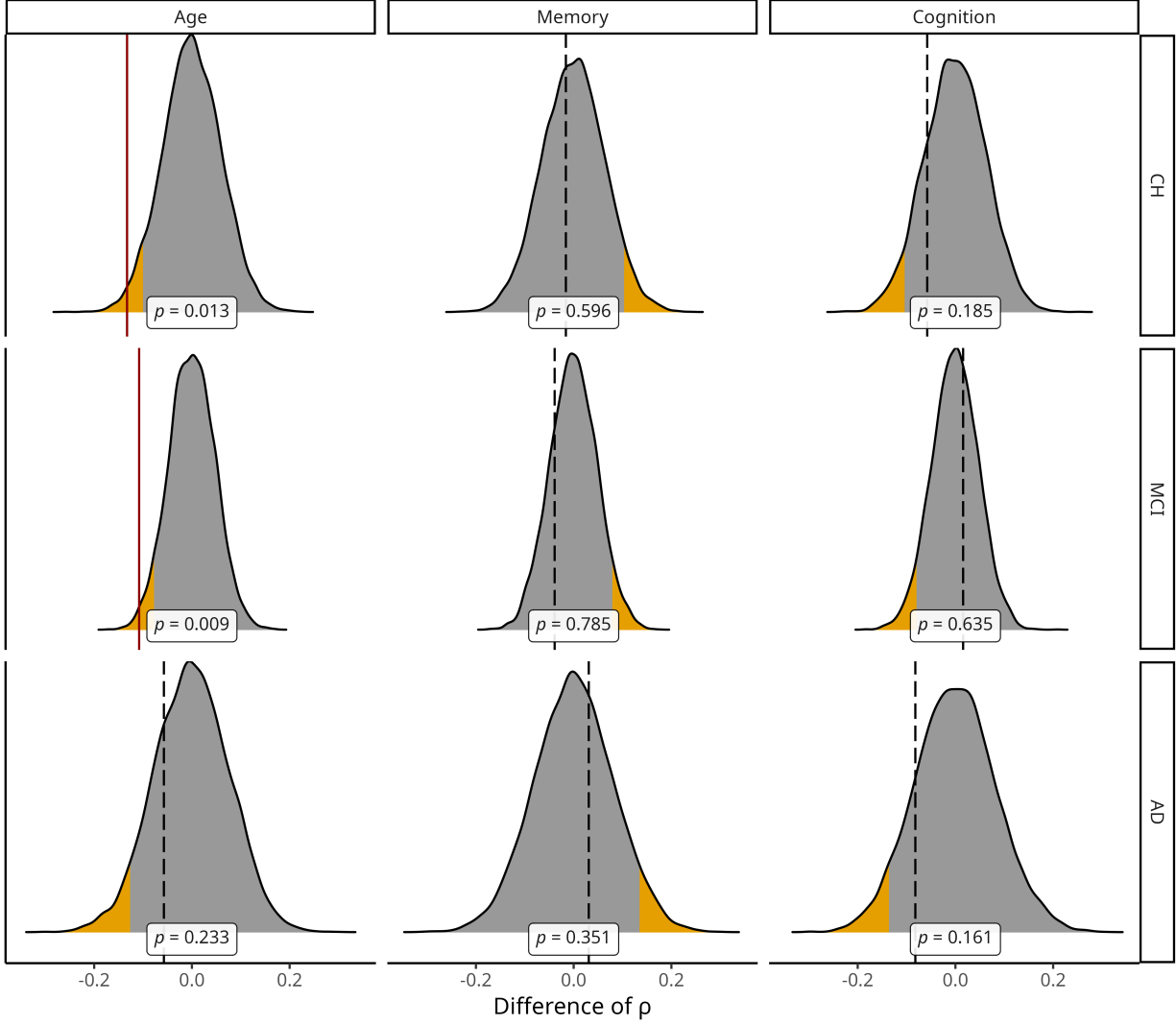


FIGURE S6 – Permutation tests for differences in correlation strength between HCvol and HVR using CNN segmentations.

Distributions of the differences in Spearman's rank correlation coefficients ($\Delta\rho = \rho(\text{HCvol}) - \rho(\text{HVR})$) with respect to age, memory, and global cognition. The hypothesis tested is that the CNN-based HVR measure yields stronger correlations in magnitude than the CNN-based HCvol measure. Each distribution was obtained via 10,000 one-tailed permutation tests; shaded orange area represents the region of rejection of the null hypothesis ($\alpha < 0.05$). Vertical lines mark the observed $\Delta\rho$; solid and red when significant, black and dashed when not. FDR-corrected p -values are provided as labels.

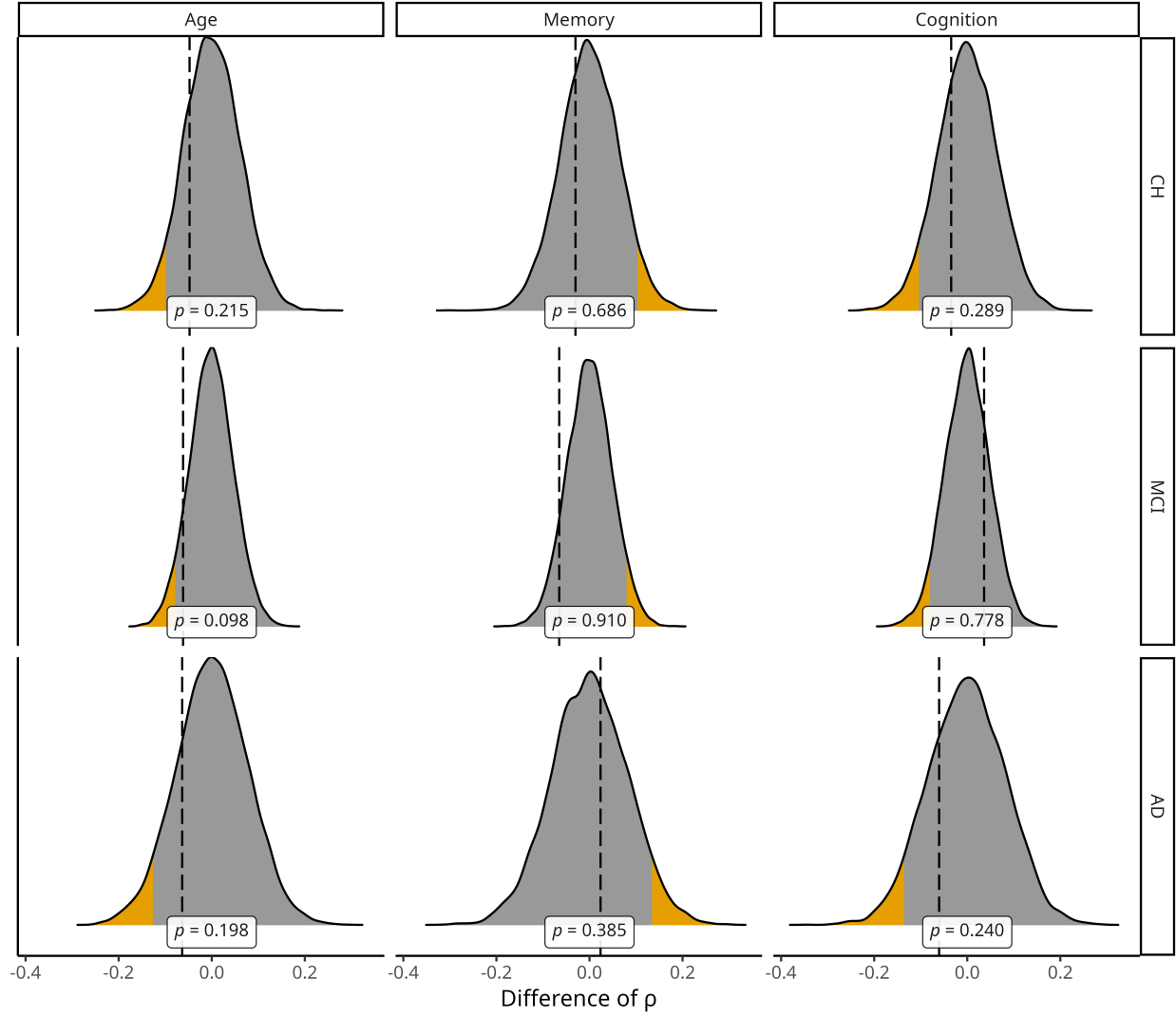


FIGURE S7 – Permutation tests for differences in correlation strength between HCvol and HVR using FreeSurfer segmentations.

Distributions of the differences in Spearman's rank correlation coefficients ($\Delta\rho = \rho(\text{HCvol}) - \rho(\text{HVR})$) with respect to age, memory, and global cognition. The hypothesis tested is that the FreeSurfer-based HVR measure yields stronger correlations in magnitude than the FreeSurfer-based HCvol measure. Each distribution was obtained via 10,000 one-tailed permutation tests; shaded orange area represents the region of rejection of the null hypothesis ($\alpha < 0.05$). Vertical lines mark the observed $\Delta\rho$; solid and red when significant, black and dashed when not.

FDR-corrected p -values are provided as labels.

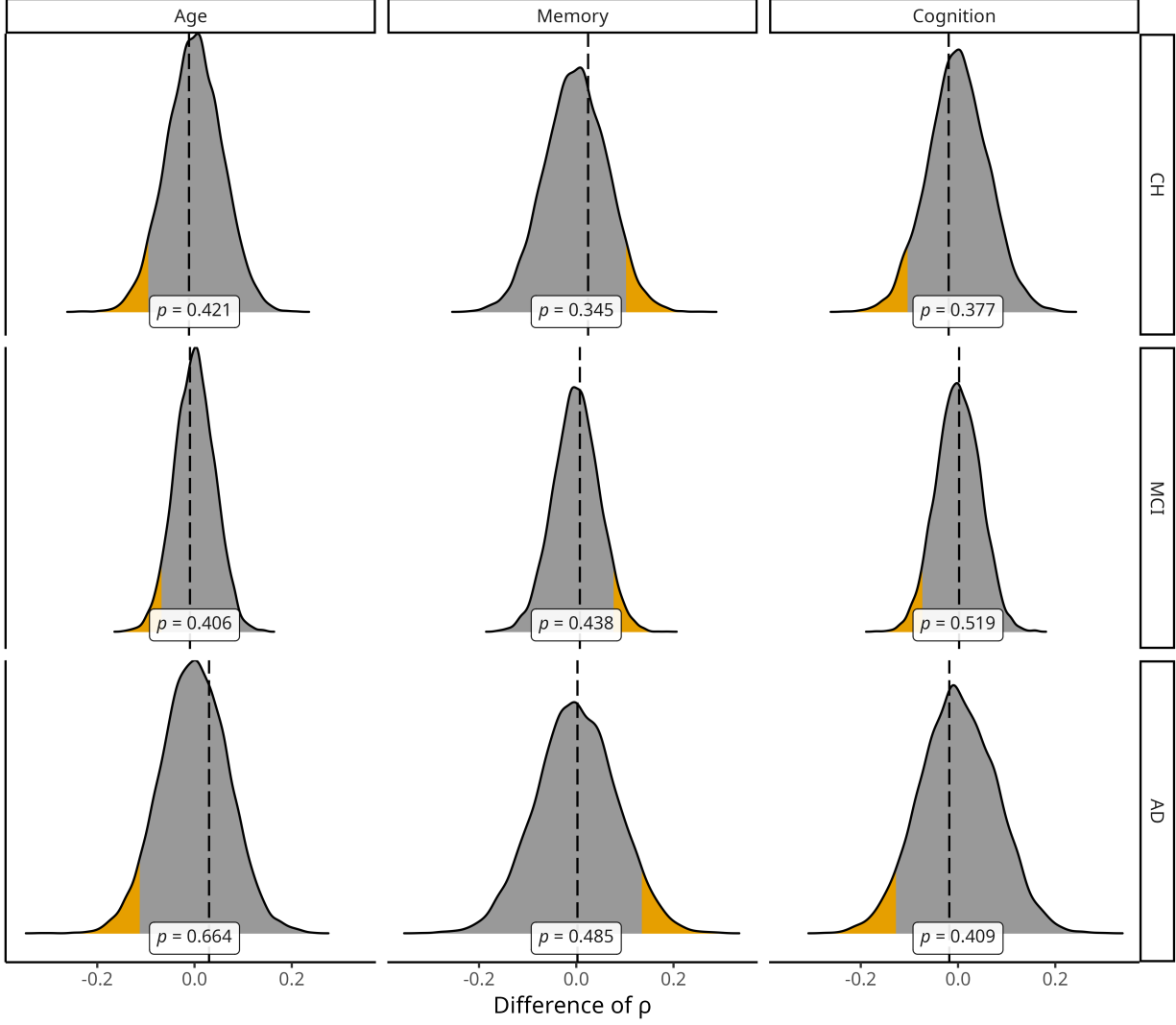


FIGURE S8 – Permutation tests for differences in correlation strength of HVR between CNN and FreeSurfer segmentations.

Distributions of the differences in Spearman's rank correlation coefficients ($\Delta\rho = \rho(\text{CNN}) - \rho(\text{FS})$) with respect to age, memory, and global cognition. The hypothesis tested is that the CNN-based HVR measure yields stronger correlations in magnitude than the FreeSurfer-based HVR measure. Each distribution was obtained via 10,000 one-tailed permutation tests; shaded orange area represents the region of rejection of the null hypothesis ($\alpha < 0.05$). Vertical dashed lines mark the observed $\Delta\rho$. FDR-corrected p -values are provided as labels.